

- Circular Convolution (for periodic DTS)

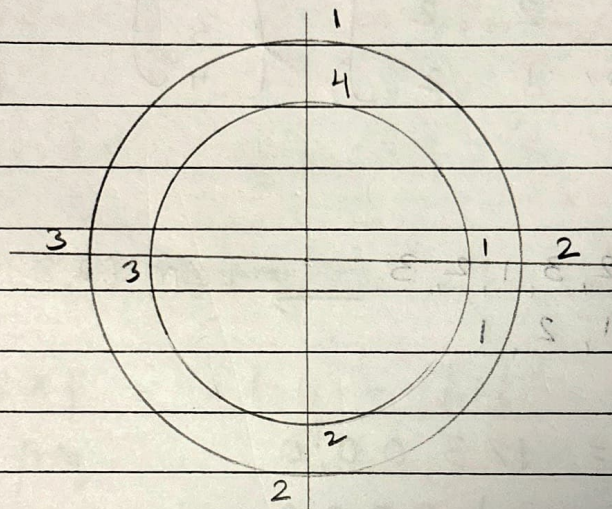
$$x[n] = \boxed{1, 2, 3, 5}, 1, 2, 3, 5 ; L_1 = 4$$

$$h[n] = \boxed{4, 5, 6, 7}, 4, 5, 6, 7 ; L_2 = 4$$

Q. $x[n] = 1, 2, 3, 4, 1, 2, 3, 4, 1, 2, 3, 4$

$$h[n] = 2, 1, 3, 2, 2, 1, 3, 2$$

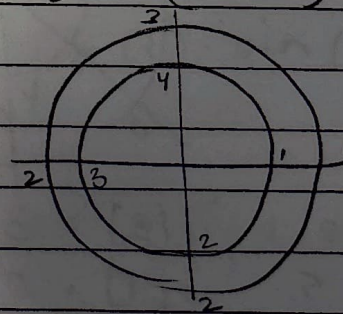
$$y[n] = x[n] \otimes h[n]$$



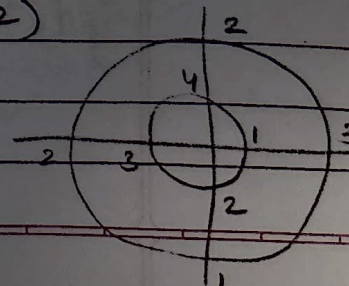
$$y[0] = (1 \times 2) + (2 \times 2) + (3 \times 3) + (4 \times 1) = 19$$

$$y[1] = (1 \times 1) + (2 \times 2) + (3 \times 2) + (4 \times 3) = 23$$

(shift by 90°)



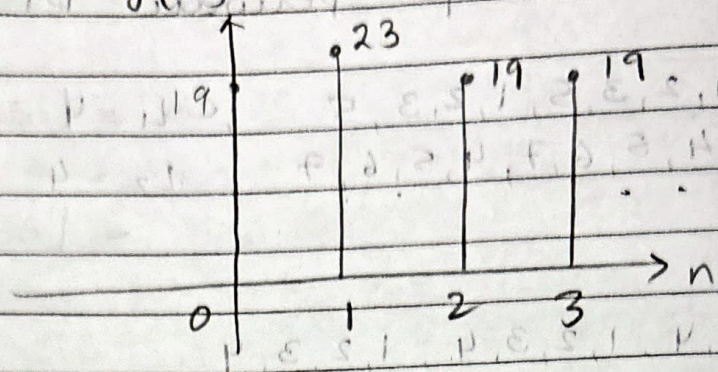
$$y[2] = (1 \times 3) + (2 \times 1) + (3 \times 2) + (4 \times 2) = 19$$



$$y[3] = 2 + 6 + 3 + 8 = 19$$

$$y[4] = y[0] = 19$$

(Convolution) $y[n]$

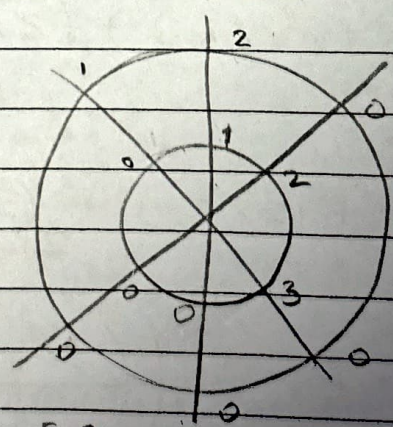


$$= \begin{bmatrix} 2 & 2 & 3 & 1 \\ 1 & 2 & 2 & 3 \\ 3 & 1 & 2 & 2 \\ 2 & 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 19 \\ 23 \\ 19 \\ 19 \end{bmatrix}$$

Q. $x[n] = 1, 2, 3, 1, 2, 3 \rightarrow \text{LCM}(2, 3) = 6 \rightarrow \text{length}$
 $h[n] = 2, 1, 2, 1$

$$x[n] = 1, 2, 3, 0, 0, 0$$

$$h[n] = 2, 1, 0, 0, 0, 0$$

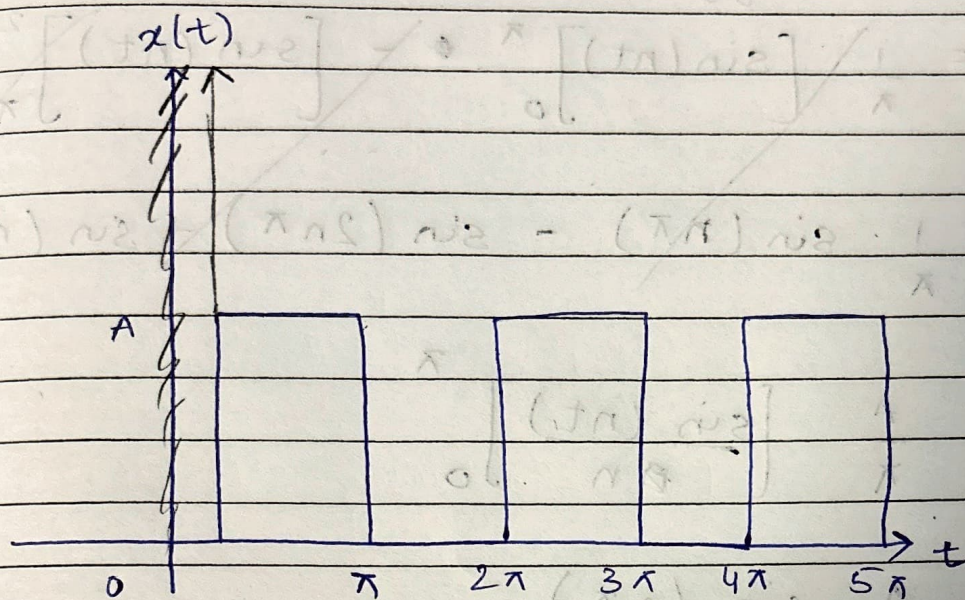


$$y[0] = 2, \quad y[3] = 3, \quad y[6] = 2$$

$$y[1] = 5, \quad y[4] = 0$$

$$y[2] = 8, \quad y[5] = 0$$

Fourier Series



End pt $\rightarrow 2\pi$ (after 2π its repeating)

$$T = 2\pi - 0 \text{ (start end pt - start pt)}$$

Time period $\leftarrow T = 2\pi$

$$\omega = 2\pi f = 2\pi \cdot \frac{1}{T} = \boxed{1}$$

↓
angular freq

Fourier coefficients

$$a_0 \text{ (overall setting)} \rightarrow \frac{1}{T} \int_0^T x(t) \cdot dt$$

$$a_n \text{ (cos)} \rightarrow \frac{2}{T} \int_0^T x(t) \cdot \cos(n\omega t) \cdot dt$$

$$b_n \text{ (sin)}$$

$$\rightarrow \frac{2}{T} \int_0^T x(t) \cdot \sin(n\omega t) dt$$

$$a_0 = \frac{1}{2\pi} \int_0^\pi A \cdot dt + \frac{2\pi}{\pi} \int_\pi^{2\pi} 0 \cdot dt$$

$$= A/2$$

$$a_n = \frac{1}{\pi} \int_0^{\pi} A \cos(nt) dt + \frac{1}{\pi} \int_{\pi}^{2\pi} 0 \cdot \cos(nt) dt$$

$$= \frac{1}{\pi} \left[\sin(nt) \right]_0^{\pi} - \left[\sin(nt) \right]_{\pi}^{2\pi}$$

$$= \frac{1}{\pi} \sin(n\pi) - \sin(2n\pi) + \sin(n\pi)$$

$$= \frac{A}{\pi} \left[\sin(nt) \right]_0^{\pi}$$

$$= \frac{A}{n\pi} \sin(n\pi)$$

$$= 0 \rightarrow \text{multiples of } \sin(\pi) = 0$$

$$b_n = \frac{A}{\pi} \int_0^{\pi} \sin(nt) dt$$

$$= \frac{A}{\pi} \left[-\frac{\cos(nt)}{n} \right]_0^{\pi} = -\frac{A}{n\pi} [\cos(n\pi) - 1]$$

$$= -\frac{A}{n\pi} [(-1)^n - 1]$$

$n - \text{even} \rightarrow 0$
 $n - \text{odd} \rightarrow \frac{2A}{n\pi}$